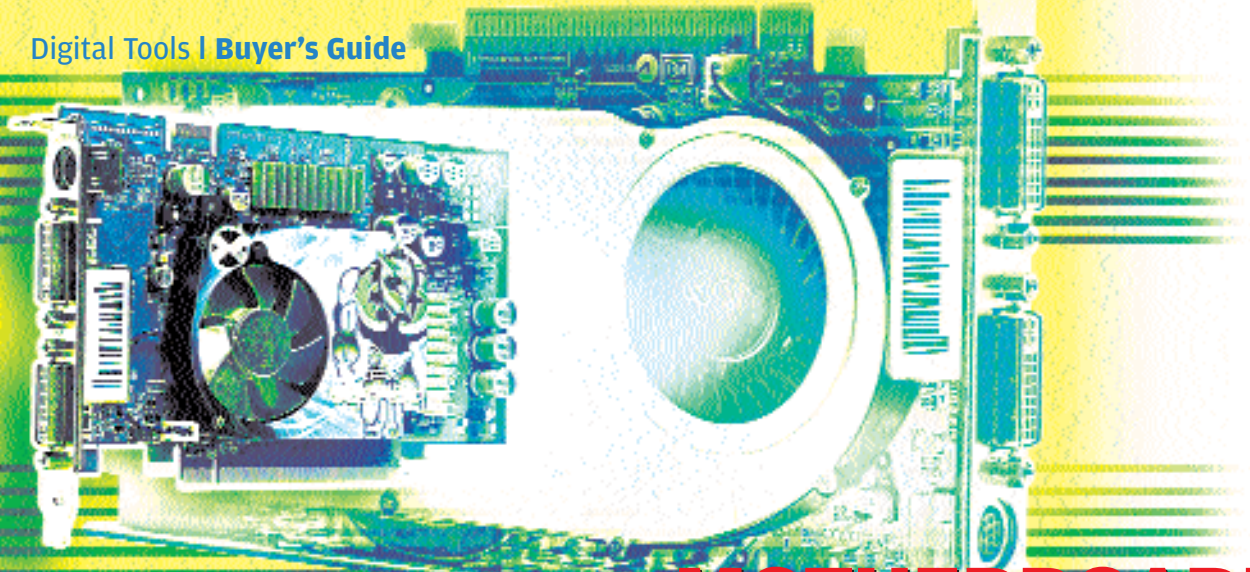




MOTHERBOARDS

The motherboard is one of the most important of all the PC components—it handles all the components and their interactions with each other. System performance depends to a large extent on the motherboard. Today, we see motherboards that come with a host of features such as RAID, SATA, WiFi, SLI, etc., to name a few. The architecture has also evolved from the slow ISA bus to the faster PCI bus and now on to the even faster PCI-Express bus.

Questions To Ask The Dealer

Does the motherboard have dual-core processor support?

This can be usually determined from the chipset on which the motherboard is based. But certain chipsets are known to partially support or require a BIOS update to support dual-core processors.

Is the motherboard graphics upgradeable?

If the motherboard you're looking at purchasing has onboard graphics, check if it has a PCI-Express graphics slot. This will enable you to upgrade to a faster PCI-

Express graphics card at a later date.

Does the motherboard have overclocking features?

While most motherboards have overclocking features of some sort, some manufacturers offer extensive overclocking features on their premium boards.

Does the motherboard support faster CPUs and memory?

You should verify that the motherboard supports faster processors than what you are currently purchasing. It should have support for memories with speeds faster than what you have right now. This is important for upgrades in the future.

Interesting Facts

Motherboards have been available in a variety of form factors designed mainly keeping the size of the computer cabinet in perspective. IBM created the PC/XT for the IBM PC in 1981, and soon it became the *de facto* standard for motherboards. Then came the AT form factor—also known as full AT—which succeeded the PC/XT, and this form

factor gained prominence during the era of the Intel 80386 processors. The ATX form factor, a miniaturisation of the AT, was next, and this was again miniaturised into Mini ATX and Micro ATX which are in use today.

The architecture of the motherboard evolved from the ISA to PCI and now to PCI-Express.



Usage Tips

❑ Check the motherboard manufacturer's Web site periodically for updated BIOS, chipset and other integrated component drivers. These updated drivers provide bug fixes and enhance system performance.

❑ If you are overclocking, use a power supply of at least 400 watts for stability.

❑ CD and DVD writers are likely to produce coasters when a system is overclocked, so you should not overclock if you need to do tasks—such as disc burning—that are dependent on the system's FSB.

❑ While mounting the processor on the motherboard socket, take care to properly align the notch on the CPU with that on the socket to avoid damage to the processor or motherboard.

❑ A motherboard needs housekeeping, and you should periodically clean it to remove the dust that gathers. This reduces accumulation of heat and static electricity, and lengthens the life of the motherboard along with that of other components. ■



Points To Ponder

Should I opt for a PCIe- or AGP-based motherboard?

AGP is all but obsolete. PCIe has almost replaced it. And if you plan to upgrade the graphics of your PC at a later date, PCIe would be your best bet.

Is the onboard graphics capable of running Windows Vista?

In order to be able to run Windows Vista, the motherboard's onboard graphics must be DirectX 9 compliant.

What form factor should the motherboard have?

If you are upgrading, buy a motherboard with the form factor that can fit inside your PC cabinet. If you have a small cabinet, a Micro ATX motherboard should be purchased rather

than a full ATX motherboard, which will not fit. It should be noted that a full ATX motherboard generally comes with better features, and is priced higher.

What type of power supply connector does the motherboard require?

There are two types of power supply connectors—20-pin and 24-pin. Make sure that you check this if you are upgrading the motherboard.

What are the frills that I should look for in a motherboard?

Buy a motherboard that has features such as FireWire, USB 2.0, SATA, RAID, and support for 64-bit processors and dual-core processors as well as dual graphics card configurations such as SLI and CrossFire.